

Aman Kumar

📞 +91-7681824426 | ✉ kaman7501@gmail.com | 🌐 aman-kumar-818412172 | 🏠 Amank7501

PROFESSIONAL SUMMARY

Full-Stack Software Developer with **2+ years** of experience building **scalable low-code platform**, **AI-assisted workflows**, and **real-time collaboration systems**, with a strong foundation in **distributed system architecture** and a proven track record of **leading teams** and delivering **high-impact platform capabilities**.

TECHNICAL SKILLS

Languages & Core: JavaScript/TypeScript, Java, C/C++, SQL

Frontend: React, HTML5, CSS3, Tailwind CSS, Bootstrap

State & UI Systems: Zustand, React Flow, React DnD, AG Grid

Backend & Real-time: Node.js, Express.js, REST APIs, WebSockets, JWT

Architecture: Distributed systems, Offline-first sync, Conflict detection & resolution, Multi-user collaboration, Low-code codegen, Workflow orchestration, Live preview, Version control

Data & DevOps: MongoDB, MySQL, Docker, GitHub, GitHub Actions, CI/CD

PROFESSIONAL EXPERIENCE

Software Developer (Full-Time)

July 2024 – Present

CTRI-Digital Governance, IIIT Bangalore 🏠

Owned end-to-end platform frontend — code generation, live preview, distributed collaboration, and AI codegen integration

- Engineered the **frontend generation module from scratch**, converting abstract UI models into production-ready **React applications** with complex component hierarchies, dynamic state management, and automatic routing
- Designed a full **code-generation pipeline** supporting dynamic component composition, style integration, and API bindings — enabling **non-technical users** to generate complete full-stack applications without writing a single line of code
- Developed a **distributed multi-user collaboration system** with offline-first sync, unit-level versioning, and hash-based conflict detection — supporting concurrent development across **10+ simultaneous users**
- Integrated **AI Codegen** for automated **Spring Boot backend generation** via workflow orchestration and resumable multi-step flows — reducing backend scaffolding time by **70%**
- Spearheaded a **drag-and-drop UI builder** using React, TypeScript, React DnD, and Zustand with real-time component manipulation and live state preview
- Implemented **live preview** functionality enabling real-time UI rendering without full application generation, improving developer iteration speed by **60%**
- Delivered a production grade **No-Dues application** for IIIT Bangalore (**nodues.iiitb.net**) **end-to-end in 2 days** using RASP — from requirements and architecture design to full-stack code generation (React + AI-generated Spring Boot backend), manual testing, and deployment — serving **300+ students** across departments
- Deployed and managed **3+ production-grade services** (RASP core, Collaboration Server, AI Codegen, Workflow Service, IR-to-Code Service) using **Docker** — handling containerization, environment config, and multi-service orchestration
- Mentored **20+ interns and junior engineers**, leading code reviews and architectural decisions across multiple teams

Software Developer (Intern)

Jan 2024 – June 2024

CTRI-Digital Governance, IIIT Bangalore 🏠

Built foundational frontend generation module and component pipeline for a model-driven low-code platform

- Engineered the core **frontend generation engine** using React.js, TypeScript, and Node.js, translating metadata-driven UI configurations into production-ready React components with integrated styling and state management
- Established **Node.js APIs** to convert abstract UI models into downloadable React application files, supporting **5+ component types** with dynamic prop binding and style injection
- Collaborated with senior engineers to architect the **component generation pipeline**, establishing extensible patterns adopted across the full platform

EDUCATION

M.Tech, Computer Science & Engineering

July 2022 – July 2024

International Institute of Information Technology, Bangalore

ACHIEVEMENTS & EXHIBITIONS

- Exhibitor, **Bangalore International Tech Summit 2025** — Represented CTRI-DG @ IIIT Bangalore and demonstrated the full scope of RASP to industry leaders, government officials, and international delegates; live demos covered the drag-and-drop UI builder, model-driven workflow engine, full-stack application generation and the complete end-to-end developer experience
- **GATE 2022** — Secured **97.96 percentile** in Computer Science and Engineering